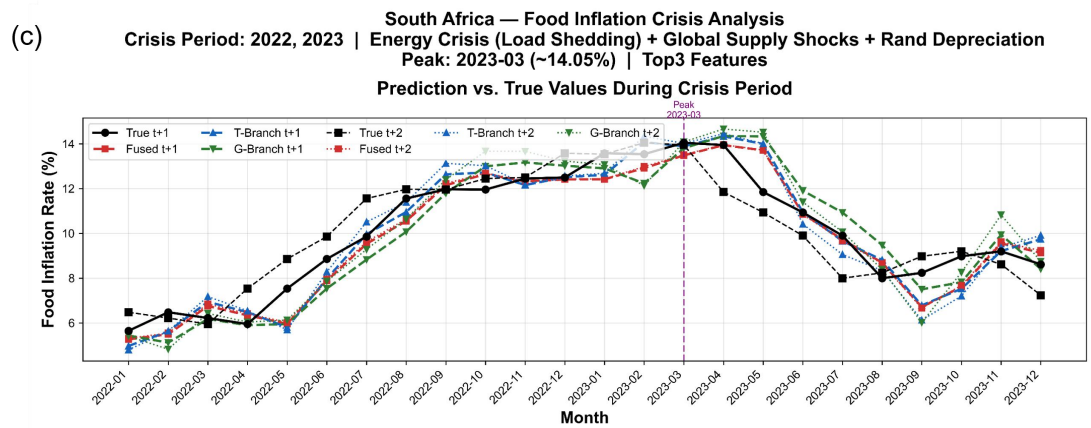
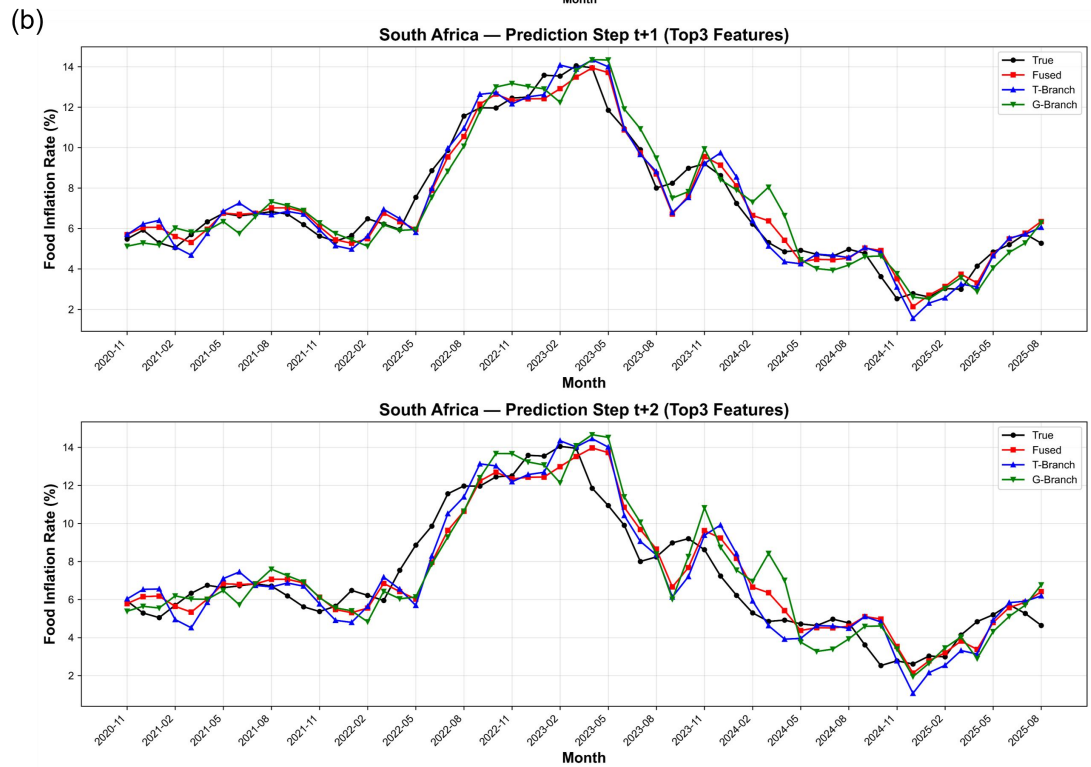
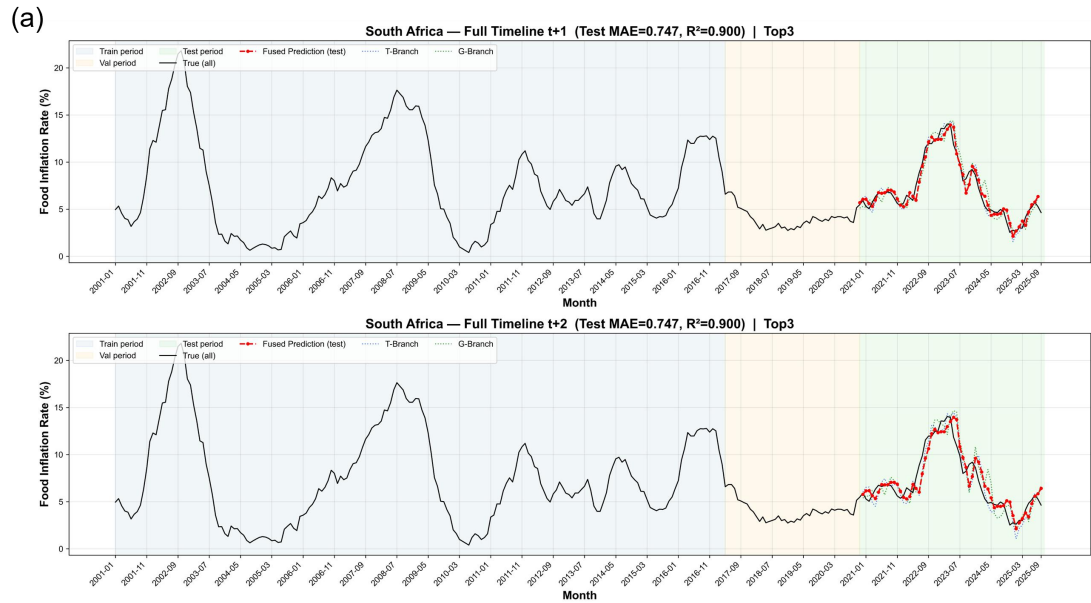
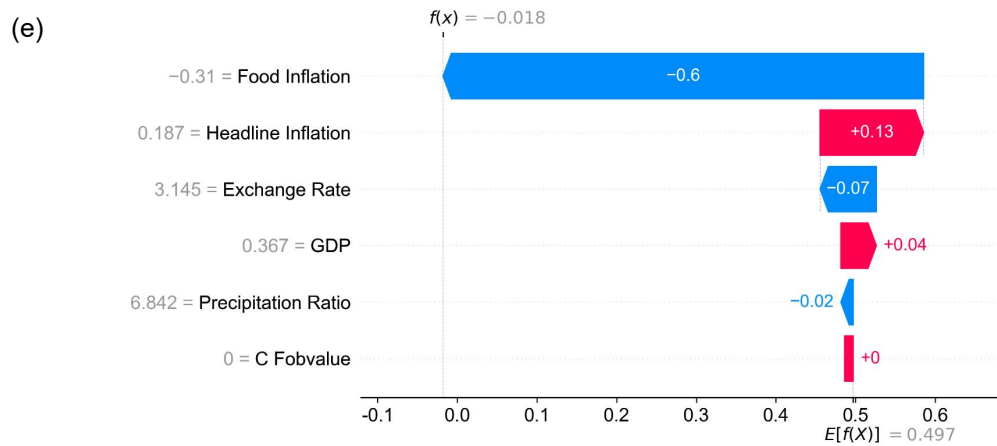
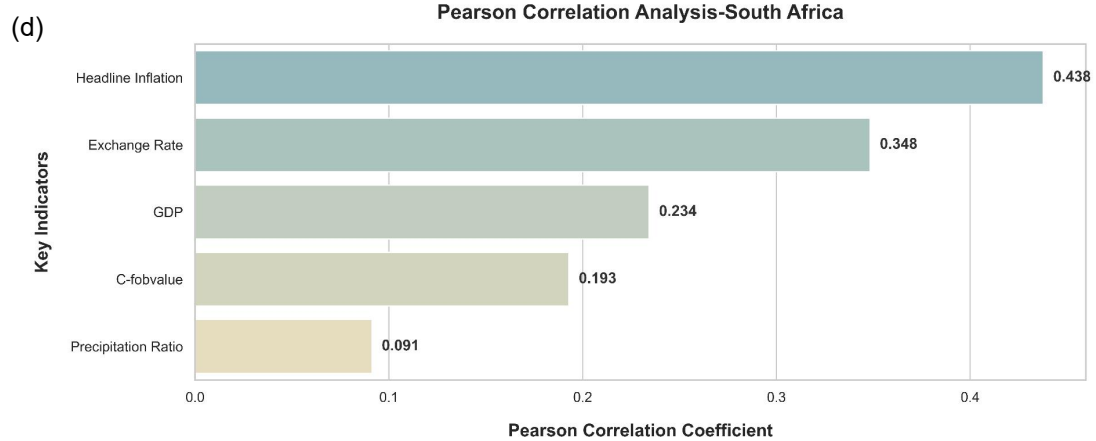






Multidimensional Evaluation and Attribution Analysis of Food Inflation Forecast in South Africa. (a) Overview of the total cycle prediction and fitting, (b) Multi-step Short-Term Forecast Comparison Chart, (c) Details of Prediction Performance in Extreme Crisis Periods, (d) Pearson correlation analysis of macroeconomic drivers, (e) SHAP-based explanation of feature contributions for a specific prediction instance.

Deeply influenced by the superimposition of domestic structural constraints alongside external macroeconomic volatility, a persistent energy deficit, a weakening exchange rate, coupled with global supply chain shocks constitute the core dominant factors driving South African food inflation. Serving as a regional breadbasket, the nation experiences rapid surges in input costs for highly import-dependent agrochemicals coupled with fuels whenever extreme climate anomalies or geopolitical conflicts escalate global commodity prices. On the production side, unprecedented frequent rolling blackouts have severely disrupted agricultural infrastructure alongside cold chain logistics, forcing enterprises to invest heavily in alternative power generation while transferring these surging operational costs entirely to consumers. This typical structural cost-push inflation ultimately allows domestic infrastructure bottlenecks alongside global price shocks to form a resonance amplification effect transmitted via the depreciating South African Rand.